

Mosier, OR - Oil Train Derailment

Oil & Gas Facility · 2026-09-30

OVERALL CONDITION CRITICAL	5 Findings	4 P1 Immediate	1 P2 Planned	0 P3 Monitor
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Source footage	mosier_oil_train.ogv
Footage length	68.2 s
Frames analysed	15 (one every 4.26 s)
Raw observations	8 (0 rejected as likely false alarms)
Detection model	Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)
Triage model	TypeSafe Jev (jev-1.13.0)
Processing time	2.7 s

How urgent is each priority

P1	Immediate (within 24-72 h)
P2	Planned (within 30 days)
P3	Scheduled / monitor (next maintenance cycle)

1. Executive summary

Aerial footage filmed by a US Coast Guard helicopter on 3 June 2016 shows an active derailment of a crude-oil unit train at Mosier, Oregon. Several derailed tank cars are burning with a dense black smoke column (F-02), about six more are jackknifed off the track beside the fire (F-02, F-03), and smoke is drifting over nearby homes (F-04). Intact tank cars on the same track sit within a few car lengths of the flames and are exposed to its heat (F-01). A pond beside the stationary train shows yellow-green streaks that could be an oil sheen but look equally like pollen or algae (F-05); it needs a water check before any conclusion. No oil could be seen on the Columbia River in the footage, but the video's low resolution (512x288) means a thin sheen could be missed.

Key risks

- F-02: Burning crude-oil tank cars; fire can spread to neighbouring cars and heat-exposed tanks can rupture.
- F-01: Intact, loaded tank cars a few car lengths from the fire; cooling them is critical.
- F-04: Smoke from burning crude drifting over homes; residents' exposure.
- F-02, F-03: About six derailed tank cars may have punctured shells or damaged valves once the fire is out.
- F-05: Possible release reaching surface water next to the rail line (unconfirmed).

Findings by category

Fire or explosion		4
Hydrocarbon leak		1

2. Action plan

Immediate (24-72 hours)

- Keep the evacuation / shelter-in-place zone around the fire and downwind homes (F-02, F-04).
- Fight the fire with foam and cool the adjacent intact tank cars with water (F-02, F-01).
- Monitor air quality in the homes under the smoke (F-04).
- Put booms and berms between the derailment and nearby water and drains to catch oil and firefighting runoff.
- Sample the pond beside the train with a sheen net or water sample (F-05).

Next 30 days

- Inspect every derailed and heat-exposed tank car for leaks; offload remaining crude to sound tanks before moving them (F-02, F-03, F-01).
- Remove the damaged cars, excavate contaminated soil and repair the track.
- Survey the Columbia River shoreline downstream on foot or by boat for sheen, since this video's resolution can't rule it out.

Next 90 days

- Complete soil and water monitoring at the site and pond, and close out regulatory reporting.
- Re-fly the site with a high-resolution drone camera to confirm cleanup and track restoration.

3. Findings register

All findings, most urgent first. Each is detailed with photos and repair steps in section 4.

ID	Time	Issue	Category	Severity	Priority	Action
F-01	00:12.8	Intact tank cars close to the fire	fire or explosion	Critical (4.0)	P1	Shut down
F-02	00:25.6	Crude oil fire with black smoke column	fire or explosion	Critical (4.0)	P1	Shut down
F-03	00:25.6	Derailed tank cars in zig-zag, several on their sides	fire or explosion	Critical (4.0)	P1	Shut down
F-04	00:25.6	Smoke drifting over nearby homes	fire or explosion	Critical (4.0)	P1	Shut down
F-05	00:04.3	Yellow-green streaks on pond next to tank-car train [verify on site]	hydrocarbon leak	Major (3.3)	P2	Monitor

4. Detailed findings

F-01 Intact tank cars close to the fire

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:12.8 (seen in 1 frame(s), from 00:12.8)
Component	undamaged tank cars on the track next to the burning section
Extent	At least 6 cars within a few car lengths of the fire
Category	fire or explosion (100% confidence)
Severity	Critical, score 3.99 / 4
Safety / environmental hazard	90% probability
Likelihood it is a real defect	87%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Emergency services + site incident command + HSE + hazmat/spill contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The next tank cars down the line are still on the rails and look intact, but they sit only a few car lengths from the fire and are exposed to its heat.

Visual evidence: Two groups of black tank cars upright on the track, the nearest roughly 2-4 car lengths from visible flames.

How to fix it

For this finding: Uncouple and pull the intact cars away from the fire once it is safe, or keep them cooled with water until they can be moved.

Standard procedure

- Emergency response first: call emergency services, activate the site emergency plan and set an evacuation or shelter-in-place zone (downwind and around tank cars/vessels).
- Stop the fuel: close isolation valves and shut down pumps feeding the fire if it can be done safely.
- Fight hydrocarbon fires with foam, not water jets; cool nearby tanks, tank cars and vessels with water to prevent them overheating and rupturing.
- Keep responders upwind; monitor air quality in nearby homes and at the site boundary.
- Contain runoff (foam, water and oil) away from drains, ditches and waterways with booms and berms.
- After extinguishment: inspect every exposed or damaged tank for punctures, valve damage and heat damage before moving product; offload to sound tanks.

7 Report the incident to the regulators that apply (e.g. National Response Center in the US) and preserve evidence for the investigation.

References: NFPA 30 (flammable and combustible liquids); NFPA 11 (foam systems); PHMSA Emergency Response Guidebook (ERG); 40 CFR Part 112 (SPCC)

F-02 Crude oil fire with black smoke column

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	00:25.6 (seen in 3 frame(s), from 00:12.8)
Component	burning tank cars at the head of the derailment
Extent	Flames across about 2-3 car lengths
Category	fire or explosion (100% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	97% probability
Likelihood it is a real defect	84%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Emergency services + site incident command + HSE + hazmat/spill contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Flames reach several metres high at the derailment and a dense black smoke column rises straight up out of the frame.

Visual evidence: Bright orange flames and a column of black smoke rising from the burning cars, with debris scattered at their base.

How to fix it

For this finding: Keep foam-equipped crews on the fire and cool the nearby tank cars with water until the fire is fully out.

Standard procedure

- Emergency response first: call emergency services, activate the site emergency plan and set an evacuation or shelter-in-place zone (downwind and around tank cars/vessels).
- Stop the fuel: close isolation valves and shut down pumps feeding the fire if it can be done safely.

- 3

Fight hydrocarbon fires with foam, not water jets; cool nearby tanks, tank cars and vessels with water to prevent them overheating and rupturing.
- 4

Keep responders upwind; monitor air quality in nearby homes and at the site boundary.
- 5

Contain runoff (foam, water and oil) away from drains, ditches and waterways with booms and berms.
- 6

After extinguishment: inspect every exposed or damaged tank for punctures, valve damage and heat damage before moving product; offload to sound tanks.
- 7

Report the incident to the regulators that apply (e.g. National Response Center in the US) and preserve evidence for the investigation.

References: NFPA 30 (flammable and combustible liquids); NFPA 11 (foam systems); PHMSA Emergency Response Guidebook (ERG); 40 CFR Part 112 (SPCC)

F-03 Derailed tank cars in zig-zag, several on their sides

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	00:25.6 (seen in 2 frame(s), from 00:21.3)
Component	derailed crude oil tank cars
Extent	About 6 cars over roughly 100 m of track
Category	fire or explosion (55% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	96% probability
Likelihood it is a real defect	96%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Emergency services + site incident command + HSE + hazmat/spill contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

About six tank cars have jackknifed into a zig-zag off the track, several lying at steep angles or on their sides. Tank cars in this position can have punctured shells, sheared valves or damaged fittings.

Visual evidence: Cylindrical tank car bodies at sharp alternating angles, some lying almost across the track, directly next to the fire.

How to fix it

For this finding: After the fire is out, check every derailed car for leaks, transfer the remaining oil to undamaged tanks, then rerail or remove the cars and repair the track.

Standard procedure

- 1 Emergency response first: call emergency services, activate the site emergency plan and set an evacuation or shelter-in-place zone (downwind and around tank cars/vessels).
- 2 Stop the fuel: close isolation valves and shut down pumps feeding the fire if it can be done safely.
- 3 Fight hydrocarbon fires with foam, not water jets; cool nearby tanks, tank cars and vessels with water to prevent them overheating and rupturing.
- 4 Keep responders upwind; monitor air quality in nearby homes and at the site boundary.
- 5 Contain runoff (foam, water and oil) away from drains, ditches and waterways with booms and berms.
- 6 After extinguishment: inspect every exposed or damaged tank for punctures, valve damage and heat damage before moving product; offload to sound tanks.
- 7 Report the incident to the regulators that apply (e.g. National Response Center in the US) and preserve evidence for the investigation.

References: NFPA 30 (flammable and combustible liquids); NFPA 11 (foam systems); PHMSA Emergency Response Guidebook (ERG); 40 CFR Part 112 (SPCC)

F-04 Smoke drifting over nearby homes

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:25.6 (seen in 1 frame(s), from 00:25.6)
Component	residential houses near the rail line
Extent	Several houses under the smoke
Category	fire or explosion (100% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	89% probability
Likelihood it is a real defect	80%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Emergency services + site incident command + HSE + hazmat/spill contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

White and grey smoke is drifting across a row of houses at the top left, a short distance from the fire. People living there may be breathing it.

Visual evidence: Roofs and houses at the top-left corner partly hidden by a white smoke haze coming from the fire.

How to fix it

For this finding: Keep the evacuation or shelter-in-place order for these homes, and monitor air quality there until the fire is out.

Standard procedure

- 1 Emergency response first: call emergency services, activate the site emergency plan and set an evacuation or shelter-in-place zone (downwind and around tank cars/vessels).
- 2 Stop the fuel: close isolation valves and shut down pumps feeding the fire if it can be done safely.
- 3 Fight hydrocarbon fires with foam, not water jets; cool nearby tanks, tank cars and vessels with water to prevent them overheating and rupturing.
- 4 Keep responders upwind; monitor air quality in nearby homes and at the site boundary.
- 5 Contain runoff (foam, water and oil) away from drains, ditches and waterways with booms and berms.
- 6 After extinguishment: inspect every exposed or damaged tank for punctures, valve damage and heat damage before moving product; offload to sound tanks.
- 7 Report the incident to the regulators that apply (e.g. National Response Center in the US) and preserve evidence for the investigation.

References: NFPA 30 (flammable and combustible liquids); NFPA 11 (foam systems); PHMSA Emergency Response Guidebook (ERG); 40 CFR Part 112 (SPCC)

F-05 Yellow-green streaks on pond next to tank-car train

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:04.3 (seen in 1 frame(s), from 00:04.3)
Component	pond surface beside rail embankment
Extent	Streaks across roughly the western half of the visible pond, tens of metres
Category	hydrocarbon leak (95% confidence)
Severity	Major, score 3.28 / 4
Safety / environmental hazard	35% probability
Likelihood it is a real defect	44% - VERIFY ON SITE before work
First action	Monitor: No intervention yet. Re-inspect on the next flight and track if it grows.
Crew	Operations + HSE + mechanical/pipefitter crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Long yellow-green streaks and patches float on the pond right next to the embankment where loaded crude tank cars are standing. They could be an oil sheen from the train, but they look equally like pollen or algae, which is common on still water in early June.

Visual evidence: Diffuse yellow-green bands on blue reflective water, no rainbow iridescence and no dark oil at the shoreline; resolution too low (512x288) to tell sheen from pollen.

How to fix it

For this finding: Have the spill response team check this pond with a sheen net or water sample before ruling out a release from the train.

Standard procedure

- 1 Treat as a potential release: notify the site supervisor and HSE lead immediately.
- 2 Isolate the affected segment (close upstream/downstream block valves) and depressurise if safe.
- 3 Deploy absorbent booms/pads; contain spread to soil or water.
- 4 Locate the source (flange gasket, valve packing, pinhole, fitting) with a ground crew and gas detector.
- 5 Repair: re-torque or replace the gasket, repack the valve, or install an engineered clamp; replace the pipe spool for through-wall corrosion.
- 6 Excavate and dispose of contaminated soil; record volume released for regulatory reporting.

References: 40 CFR Part 112 (SPCC); API 570 Piping Inspection Code; API RP 1160 (pipelines)

5. Methodology and limitations

How this report was produced

- Frames were sampled from the footage (one every 4.26 s, 15 in total).
- Detection: each frame was examined by a vision-language model (Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)) that listed every visible defect with its location, extent and visual evidence. It did not assign severity.
- Triage: each observation was scored by TypeSafe Jev (jev-1.13.0), a System One model that returns typed answers with probabilities: defect category, severity on a 5-level rubric, first action, the probability that the defect is genuine, and the probability of an immediate safety or environmental hazard.
- Rules applied by code: observations with less than 35% probability of being genuine were dropped; P1 if expected severity is 3.5 or higher or hazard probability is 60% or higher, or the recommended action is shutdown and severity is 2.5 or higher; P2 if severity is 1.75 or higher; otherwise P3. Findings with less than 60% probability of being genuine are capped at P2 and marked 'verify on site'.
- The same defect seen in neighbouring frames was merged into one finding.
- Repair procedures come from a reviewed playbook for this asset type; the per-finding fix is model-generated.

Severity scale

- 0: Cosmetic: no functional impact, note for records only.
- 1: Minor: early-stage wear; no loss of function; fix at next scheduled maintenance.
- 2: Moderate: measurable degradation or loss of output; fix within 30-90 days.
- 3: Major: active damage or significant loss of function; fix within 7-30 days.
- 4: Critical: active leak, fire, structural failure risk or danger to people or environment; act immediately.

Limitations

- Only what is visible from the air is assessed. Internal corrosion, wall thickness, electrical faults and gas leaks without a visible sign cannot be detected from RGB footage.
- Hotspot findings need radiometric thermal footage; on RGB footage they are not reliable.
- Locations are approximate boxes on the frame, not GPS coordinates.
- AI findings can be wrong. Every P1 and P2 finding should be confirmed on site before work starts.

Image credits

- Footage: "Coast Guard aircrew conducts overflight near oil train derailment", PO1 Levi Read, US Coast Guard 13th District, 3 June 2016. Public domain, via Wikimedia Commons.